ARKAIN

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ARKAIN GAMES INDIA PVT. LTD.

Junior Unity Developer Assignment (Slot Game)

Company: Arkain Games India Pvt. Ltd.
Role: Junior Unity Game Developer
Assignment Title: “Elemental Slots Lite” – 5x4 Slot Game Prototype
Duration: 3 Days

Objective

Develop a 5×4 reel slot game prototype themed around the four elements – Fire, Water, Earth, and Air. This assignment assesses your ability to implement a complete slot game loop, use structured scripts, and log simple gameplay analytics in Unity. Use any assets available for sprites of the symbols.

Game Requirements

Game Setup

- 5 reels × 4 rows (standard grid)
- 13 Symbols total:
    - 10 Regular Symbols (Element icons or placeholders)
    - 1 Wild Symbol
    - 1 Scatter Symbol
    - 1 Bonus Symbol (optional)
- 1024-ways win evaluation or a simple payline-based check.
- Basic spin button triggering a full spin cycle and result display.
- When the result is displayed it should animate the winning combination either by scale up and scale down animation or a symbol sequence png animation.
- Log the **RTP** to the Console after every spin. The **RTP** value should be between 95-96%.

Core Features

1. Random symbol generation each spin.
2. Evaluate win or loss and display result on UI.

3. Display credit balance, bet amount, and win text.
4. Log gameplay events to the Unity Console for debugging.
5. Clean modular scripts with comments and separation of concerns.

Bonus Feature (Pick One)

- **Free Spins:** Triggered by 3 Scatter symbols (auto-play 3 spins).
- **Expanding Wilds:** Wild expands vertically if it lands in the middle reel.

Code Structure

Use a modular approach with the following suggested scripts:

- **GameController.cs** — Handles main game loop, spin logic, and evaluation.
- **ReelController.cs** — Populates reels and manages spin results.
- **UIController.cs** — Manages button states, credit display, and win text.
- **AnalyticsManager.cs** — Handles event logging.
- **EventSystem.cs** — Unity's built-in Event System should be used for UI interactions.

Game Analytics (Simple Logging)

Implement a lightweight event logging system that captures major gameplay moments. No actual backend integration is required — print JSON-formatted strings to the Console. Example JSON output:

```JSON
{ "event": "spin_result", "win": 200, "symbols": ["Fire", "Earth", "Water", "Wild", "Air"] }
```

Recommended Events to Track

- **GameStart** — Triggered once when the game scene loads.
- **SpinStart** — When player presses the Spin button.
- **SpinResult** — After all reels stop spinning (include win/lose info).
- **BonusTriggered** — When Scatter/Wild condition activates a bonus.
- **BonusComplete** — After free spins or special feature ends.

- **BetChanged** — When player changes bet value.
- **SessionEnd** — When the player quits or exits.

Event System Usage

Use Unity's built-in Event System for button clicks and UI communication. For inter-script communication, you may implement a simple C# event or delegate pattern. Example:

```
C#
public static event Action OnSpinStart;
public static event Action<SpinResultData> OnSpinResult;
public static event Action OnBonusTriggered;
```

Deliverables

- **Playable Build:** Windows build with exe file to run the game.
- **Documentation:** Developer Notes or README.md describing setup, code flow, and chosen bonus logic.
- **Source Code:** Project Folder or Git Repository with organized scripts.

Demo Video (Optional): Short demo video showing gameplay and features.